Hydroponics Monitoring System

System Redesign Specification

Revision 5 | 16 October 2026 | Current design baseline

1 Project purpose and goals

The project converts the inherited hydroponics prototype into an integrated monitoring system built around a custom ESP32-S3 PCB. It measures the reservoir and surrounding air, presents trustworthy data locally and remotely, and maintains defined operation when external power or network services are unavailable.

The redesign combines a 3S battery-backed electrical system, scheduled FreeRTOS firmware, a Python application for BLE commissioning, and a network dashboard. Local alarms and one relay channel support the installation where their operating rules have been configured. Monitoring, configuration, and diagnostics remain the main project deliverables.

Project goals

G1 Deliver an integrated controller. Produce the schematic, PCB, and assembled controller for the ESP32-S3-WROOM-1-N16R8, with defined sensor connections, programming access, power distribution, and test points.

G2 Complete the measurement system. Integrate BME280, DS18B20, DFR0026, SGP30, HC-SR04, pH-4502C, and SEN0244 sensing. Each reported measurement must carry units, age, and quality; calibrated outputs require recorded reference checks.

G3 Preserve essential monitoring during an outage. Maintain the controller, BLE, and the four sensors on the retained 3.3V branch. Make the loss of external-only sensors and relay availability visible to the operator.

G4 Make firmware timing and ownership predictable. Separate acquisition, local control, power coordination, configuration, and communications. Bound waits and memory use so a slow peripheral or failed broker cannot block the rest of the system.

G5 Commission without editing firmware. Use the Python BLE application to configure network access, node identity, supported devices, and maintenance settings, with persistent storage and explicit operation results.

G6 Provide useful remote monitoring. Deliver live values, historical trends, alarms, calibration status, and node health through the backend and dashboard.

G7 Demonstrate the integrated design. Complete electrical, sensor, timing, outage, communications, and endurance tests, with results tied to the final hardware and firmware revisions.

2 System architecture

The controller owns measurement and local decisions. BLE provides a direct maintenance connection to a laptop; Wi-Fi and MQTT connect the controller to the backend for routine remote monitoring. Both communications paths consume copied measurement records and submit validated requests to the responsible firmware service. [1-3]

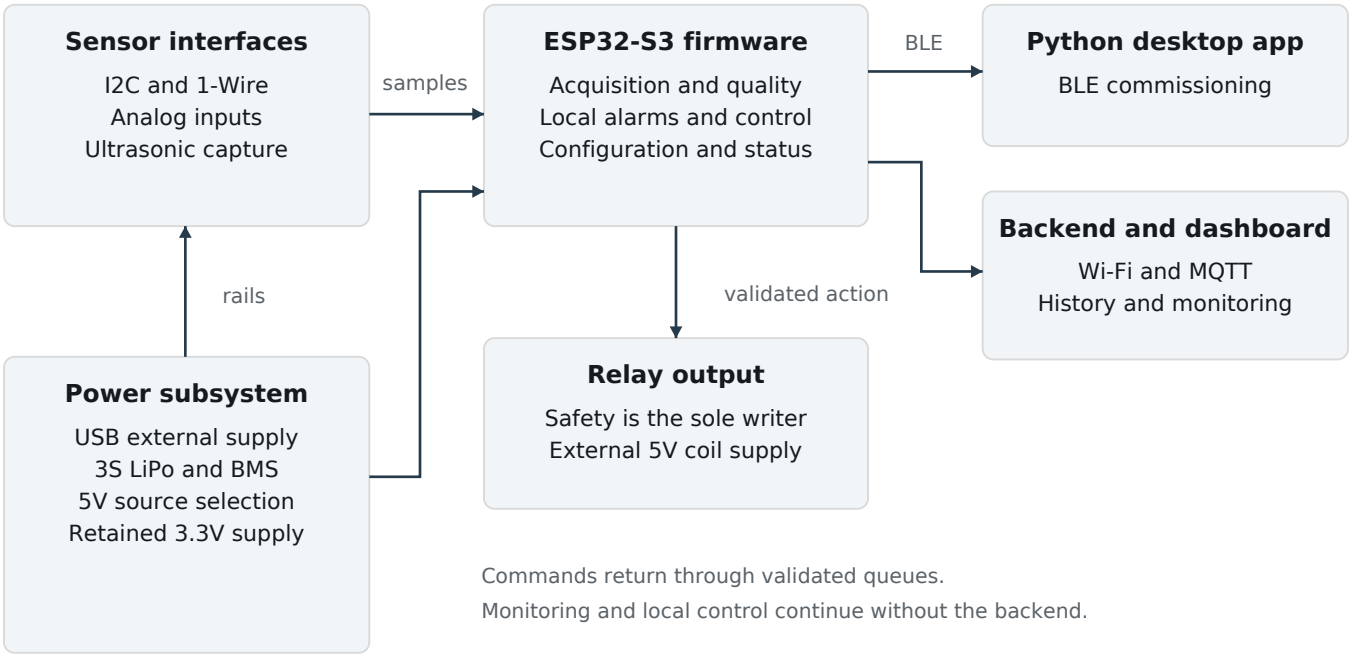


Figure 1 Functional architecture and principal data paths

Hardware baseline

The controller is the ESP32-S3-WROOM-1-N16R8. The build must configure its 16 MB flash and 8 MB PSRAM explicitly. The board uses 3.3V logic, ADC1 analog inputs, shared I2C, a dedicated 1-Wire connection, and native USB for programming and diagnostics. [2, 5]

Availability follows the electrical branches

Branch	Connected equipment	Battery operation
Retained 3.3V	ESP32-S3, BME280, DS18B20, DFR0026, SGP30 module	Available
External 5V	HC-SR04 and SRD-05VDC relay coil	Unavailable
Filtered external 5V	pH-4502C and SEN0244 TDS	Unavailable

A supply outage therefore reduces the available measurements. It does not stop all monitoring. The firmware, desktop application, and dashboard use this same availability model, including the SGP30 on the retained branch.

3 Measurement hardware and signal assignments

The board profile is the firmware contract for the signal map below. GPIO numbers identify ESP32-S3 signals, not module pad numbers. The assignments and front-end values follow the current pin drawing. [2]

Device or function	Supply	GPIO and interface	Interface treatment
BME280 Air conditions	3.3V	SDA 21, SCL 47 I2C address 0x76	Shared bus; 4.7 kΩ pull-ups to 3.3V
SGP30 module TVOC and CO2 equivalent	3.3V module input	SDA 21, SCL 47 I2C address 0x58	Shared bus; module must support the drawn supply and logic levels
DS18B20 Water temperature	3.3V	DQ 14 1-Wire	External power; 4.7 kΩ data pull-up
DFR0026 Ambient light	3.3V	1 / ADC1_CH0	10 kΩ series resistor; 1 µF to ground
SEN0244 TDS estimate	Filtered 5V	6 / ADC1_CH5	Analog input; 100 nF to ground
pH-4502C Reservoir pH	Filtered 5V	7 / ADC1_CH6	PO through 10 kΩ / 15 kΩ divider; 1 µF to ground; DO and TO unused
HC-SR04 Water level	External 5V	TRIG 17 ECHO 18	ECHO through 1 kΩ / 2 kΩ divider
External power present	External 5V signal	8 / ADC1_CH7	220 Ω upper resistor; two 220 Ω lower resistors in series
Relay drive	External 5V coil	15 / digital output	IRLZ44N; 220 Ω gate resistor; 10 kΩ pull-down; 1N5822 flyback diode
Native USB	USB interface	D− 19, D+ 20	22 Ω series resistors

Measurement meaning

Report air and water temperature in °C, humidity in % RH, pressure in hPa, and reservoir distance or level in a declared metric unit. Report pH as pH and TDS as the module-derived ppm estimate. Light is reported as raw counts or millivolts until a reference calibration supports lux.

The SGP30 reports TVOC in ppb and CO2 equivalent in ppm. CO2 equivalent is an estimated air-quality signal and must be labelled accordingly. The SGP30 IC itself uses a nominal 1.8V supply; the 3.3V connection in this design refers to a compatible sensor module. [6]

GPIO35-37 remain reserved for the N16R8 PSRAM. Other assignments must respect the S3 boot and USB functions. [2, 5]

4 Electrical power architecture

The electrical system uses separate charging and operating feeds. A protected 3S LiPo path supports the retained controller supply during an outage, while the 5V peripheral branches follow external power. These branches define the firmware power modes. [1]

Stage	Current configuration
External source	65 W multiport charger, operated at the drawn 5V outputs
Charging feed	5V / 3 A to the CN3303 charging module
Battery and protection	3S LiPo, 11.1V nominal, 5200 mAh, pack label 35C; HX-3S FL25A-A BMS; 2 A fuse in the protected load path
Operating feed	5V / 2 A through the Micro-USB input module
Backup and source selection	MP1584 set to 5V; IRLZ44N switching stage; 1N5822 isolation paths
Retained supply	Selected source bus feeds a second MP1584 set to 3.3V

Charging and protection

The CN3303 module charges the 3S pack through the BMS connections. For a conventional 4.2V-per-cell pack, the full-charge target is 12.6V; record the actual module setting and pack limits during commissioning. Charger current, BMS trip limits, and balancing capability are properties of the fitted modules and are recorded from their documentation and measurements.

The 65 W adapter rating is separate from the power available at either 5V port. Verify the simultaneous operating and charging load using the selected ports and cables. Use the selected DC-rated 2 A fuse in the battery load path.

Source selection and distribution

The operating feed supplies the external-only branches and an isolation path to the retained source bus. The battery feeds the other path through the 5V MP1584 and IRLZ44N stage. External power controls this stage; the backup path is intended to maintain the bus when the operating feed is removed. Validate source transfer, voltage drop, and startup behavior.

The source bus feeds the 3.3V MP1584. Its nominal 5V is reduced by path losses. Verify that the regulator input remains within its specified range, beginning at 4.5V, during transfer and radio current bursts. [10]

Filtering and physical layout

The pH and TDS modules share a ferrite-filtered external 5V branch with 100 nF, 1 µF, and 10 µF decoupling. Other branches use the capacitors shown in the power diagram. Place local decoupling near each load and keep relay and converter return currents away from analog returns.

5 Embedded firmware organisation

The firmware is a C++ application built with PlatformIO and the Espressif FreeRTOS kernel. Sensor drivers, application services, and communications have separate responsibilities. The ESP32-S3 board profile and resolved framework versions are recorded with the build. [3, 9]

Software layers

Drivers and hardware abstraction perform bounded hardware operations. Each sensor has its own driver with start, poll, read, cancel, and power-down operations.

Domain functions and services own scheduling, conversion, quality, alarms, power, and configuration. Each hardware resource has one owner.

Transports serialize copied records and queue commands. The entry point sets safe outputs, loads configuration, creates services, and starts the system.

Task allocation

Initial application priorities are shown below; larger numbers indicate higher priority. Verify SDK task affinity and timing in the resolved ESP32-S3 build. [3]

Task	Core	Priority	Responsibility
Safety	0	6	Sole relay writer; alarms, interlocks, command expiry, and action timers
PowerCoordinator	0	5	Power state, rail validity, and acquisition ownership
AcquisitionNormal	1	4	All qualified sensors in normal mode
AcquisitionLowPower	0	4	Retained sensors in battery mode; same acquisition engine
WiFiMqtt	0	2	Connection state, MQTT client, telemetry, and bounded retries
BLETransport	0	2	Application GATT transmission and queued command handling
Configuration	0	2	Sole device-registry and persistent-configuration writer
Supervisor	0	1	Progress, queue usage, memory, stack, and reset diagnostics
SerialLogger	1	1	Bounded logging in normal mode; blocked in battery mode

Only one acquisition worker uses the sensor buses; the other blocks. Driver development starts with one worker. Add the second worker with power-mode handover. Local actions never wait for communications or flash writes.

Measure Safety latency under concurrent Wi-Fi/BLE traffic. If higher-priority SDK work prevents the target response, revise the allocation and retest.

6 Acquisition and measurement quality

Scheduling

The active acquisition worker uses an absolute one-second frame. BME280 sampling runs every 250 ms in both operating modes; SGP30 air-quality commands retain their required one-second cadence. DS18B20, light, and the available 5V sensors are scheduled within the same engine. Communications update periods are independent of sensor periods. [3, 6]

Transactions are sequential, while internal sensor conversions may overlap. Start a conversion, yield, service another due operation, and collect the result later. The DS18B20 can take up to 750 ms at 12-bit resolution; that interval must not become a blocking delay in the acquisition task. [8]

Use an initial 30 ms HC-SR04 capture timeout. BME280 forced mode starts at ×1 oversampling without IIR delay, with a 12 ms conversion/read budget. Spread analog filter taps across their configured windows and initialize buffers with real samples. Recalculate the frame budget for the current sensor set. [3, 7]

Conversion and calibration

Retain raw analog counts and calibrated ADC millivolts. Correct pH voltage using the measured divider ratio before applying buffer calibration. SEN0244 uses a validated filter and fresh DS18B20 temperature for compensation. A failed dependency produces raw data or an explicitly uncompensated estimate. [2, 3, 11]

Initialize the SGP30 after startup or reset, check response CRCs, and mark its first 15 seconds of fixed initialization outputs as WARMING. Store baseline and compensation status separately from the readings. When humidity compensation is enabled, derive absolute humidity from fresh BME280 measurements and report dependency loss. [6]

Data contract

Each record contains sensor and metric IDs, value, unit, acquisition times, last-good time, sequence, power epoch, configuration and calibration revisions, and quality. Ages use monotonic time. Initial stale limits are 1 s for BME280 and 3 s for ordinary 1 Hz readings, subject to startup and dependency rules. [3]

Quality	Required interpretation
OK	Fresh, valid, and qualified for the configured use
STARTING or WARMING	Initialization, conversion, or filter settling is incomplete
STALE	The last valid sample has exceeded its permitted age
FAULT or TIMEOUT	A transaction, CRC, range, or deadline check failed
UNPOWERED	The sensor supply is unavailable or changed during acquisition
UNCALIBRATED or DEPENDENCY_INVALID	Physical conversion or compensation is not qualified

7 Power modes and coordinated transitions

GPIO8 senses external operating power through the drawn divider, nominally 3.33V from 5.00V. It does not measure battery voltage. Verify its actual levels and timing relative to sensor-rail decay. [1, 2]

Function	Normal mode	Battery mode
Acquisition	Core 1; all qualified sensors	Core 0; BME280, DS18B20, DFR0026, SGP30
Wi-Fi and MQTT	Enabled; reconnect independently	Stopped
BLE	Available	Available
External 5V sensors	Available after individual startup checks	UNPOWERED
Relay coil	Available from external 5V	De-energized by loss of coil supply
Core 1 application work	Acquisition and logging	Application tasks blocked

Loss of external power

A power-sense edge increments a rail epoch counter, invalidates external-rail measurements, and notifies PowerCoordinator. The interrupt performs no sensor access, printing, allocation, or networking. Invalidation is immediate; mode debounce follows it.

PowerCoordinator requests acquisition to stop new work, finish or cancel bounded operations, make interfaces safe, and release ownership. It also requests Wi-Fi/MQTT shutdown and applies relay availability policy. The retained-sensor worker receives bus access only after release.

Handover acknowledgements carry a generation number; stale or missing replies cannot create a second owner. Report TRANSITION or DEGRADED until completion. Initial targets are 50 ms loss debounce, 200 ms worker acknowledgement, and 2 s network shutdown. [3]

Return of external power

Require 500 ms of stable external power and verified rail settling. Quiesce the battery-mode worker, transfer ownership, and reinitialize restored sensors. Each remains WARMING until valid. Wi-Fi reconnects asynchronously. A further power loss aborts restoration. [3]

Reject measurements across power changes

A measurement captures rail validity and epoch at its start and checks both again at completion. Accept it only if the rail is still valid and the epoch is unchanged. This rejects a power-loss-and-return event within one acquisition window. Consumers also check the current epoch before using an external-rail value.

Battery mode remains awake for BLE and sensing. Blocking core 1 application tasks does not power off the core. Measure whole-board current and runtime.

8 Commissioning and remote monitoring

Python BLE application

The laptop application discovers controllers and opens an authenticated encrypted BLE connection. It configures network access, node identity, supported devices, and calibration metadata, and displays measurements, quality, age, mode, alarms, and results. [3, 4]

A physical procedure enables a defined commissioning window, with owner recovery. Passwords are write-only and excluded from logs. Device management validates supported types, predeclared board slots, pin conflicts, ranges, and identities. Supported instances need no reflashing; a new driver type requires firmware development.

Retain existing GATT identifiers, request IDs, and two-byte little-endian framing with a 4096-byte maximum JSON payload. Coordinate schema changes for SGP30, all BME280 metrics, quality, and rail availability. Initial subscribed update intervals are 2 s for state and 3 s for telemetry. [3]

Command and configuration handling

BLE callbacks copy bounded input and return promptly. Queued requests carry identity, request ID, session generation, and expiry where applicable. Results distinguish validation failure, busy, accepted, and applied.

Validate, stage, and persist configuration as a versioned record, then apply it at a safe boundary. Restart from the last valid commit. Driver objects change only through their owner. Rate-limit flash writes and test their timing effects.

Wi-Fi and MQTT

WiFiMqtt owns the network and MQTT client. Use finite timeouts and backoff, initially 1, 2, 4, and 8 seconds up to 60 seconds with jitter. Network loss leaves acquisition, BLE, and local rules running. Power-stop requests interrupt retries. [3]

Telemetry includes node identity, time, age, sequence, units, quality, configuration revision, mode, and diagnostics. A bounded RAM backlog preserves timestamps and reports dropped history. Never replay expired actuator commands. Remote control requires authenticated and authorized access.

Backend and dashboard

The backend ingests MQTT data, stores history and events, and serves the dashboard. The inherited gateway and database may be reused. Show live values, trends, alarms, connectivity, power mode, availability, and calibration status.

Unavailable and stale channels appear explicitly. An old reading is not displayed as a fresh observation. Battery percentage is included only after a real measurement and estimation method exists. Manual controls show the controller acknowledgement and current interlock state.

9 Local control and PCB integration

Local alarm and relay policy

Acquisition evaluates the configured thresholds immediately after accepting each valid sample. Critical events set latched pending-alarm state and notify Safety without waiting for the frame boundary or network publication. Safety is the only task that writes GPIO15. [2, 3]

Safety handles power validity and pending critical events before ordinary commands. It also checks freshness, missing dependencies, command expiry, and maximum-on duration at a bounded interval, initially 100 ms. Remote commands pass through the same local interlocks as other control requests.

Each rule defines its metric, acceptable quality, trip and clear limits, persistence time, severity, latch behavior, output action, and reset conditions. The selected installation supplies these values; example temperatures in earlier planning material are not operating limits.

The board contains one SRD-05VDC relay driven through an IRLZ44N, with a gate pull-down and flyback diode. The GPIO drive stage is active high as drawn. Verify coil response and the chosen contact wiring during commissioning. Boot and reset keep the coil de-energized. The final contact state required by the controlled load is recorded before automatic operation is enabled.

The relay coil uses external 5V and becomes unavailable during an outage. The switched load, its supply, and its required outage behavior must be documented together. A latched environmental alarm remains latched when its sensor loses power; a collapsing sensor output is treated as unavailable data.

Queues and fault handling

The latest-snapshot channel holds one copied record set and overwrites the previous version. BLE, MQTT, and Safety read it without waiting. Critical pending-alarm state survives a full detail queue; command queues return busy when full. Logs and offline history have fixed capacities and visible loss counters. [3]

Missing I2C devices, DS18B20 CRC failures, analog saturation, and capture timeouts produce explicit channel status and bounded later retries. Progress monitoring detects a stalled service independently of CPU idle time. Watchdogs and recorded reset causes support recovery; routine network loss does not restart the controller.

Custom PCB requirements

The PCB integrates the current rail topology, board profile, sensor interfaces, relay driver, reset and boot controls, and USB programming path. It respects the module antenna keep-out and reserves PSRAM pins. Connector labels show signal, supply, polarity, and orientation. [1, 2, 5]

Provide accessible test points for external 5V, the selected source bus, filtered analog 5V, retained 3.3V, protected battery voltage, GPIO8 sensing, and ground. Keep switching and relay currents away from analog routes. Place filtering and decoupling at the relevant loads, preserve a continuous ground reference, and size traces and connectors for the measured operating currents.

10 Verification and project completion

Acceptance requires measurements from the assembled controller and the final software build. The numerical values below are engineering targets carried from the firmware baseline; test records establish whether they are met on the ESP32-S3 design. [3]

Test area	Required evidence or acceptance target
Power system	Record simultaneous operating and charging current, charge endpoint, protection behavior, regulator settings, rail ripple, and temperature rise. Remove and restore the operating feed under radio load; the controller must continue without a reset or brownout.
Sensor integration	Verify device identity, startup, missing-device and timeout behavior, and measured reference error. Check both modes for every retained sensor. Qualify the SGP30 module interface and one-second command cadence.
Water and light calibration	Record pH buffer fit and an independent check point; TDS reference and temperature compensation; tank empty/full geometry; light raw response or reference lux calibration.
Schedule and local action	Trace the one-second frame and 250 ms BME280 schedule. Initial release-jitter target: ≤ 5 ms. Sample acceptance to Safety action: ≤ 10 ms under concurrent BLE/MQTT traffic. Sustained BME280 threshold detection/action target: ≤ 300 ms software delay; characterize thermal and mechanical delays separately.
Power-mode transitions	Inject loss and return at each driver stage and during pending conversions. Reject obsolete epochs; never permit two bus owners. Verify SGP30 continuity, Wi-Fi shutdown, BLE availability, restored-sensor warm-up, and relay behavior.
Communications and overload	Run 30 min with the access point or broker absent. Exercise BLE reconnects, congested frames, full command and alarm queues, credential changes, and unauthorized writes. Sensing continues and storage remains bounded.
Fault recovery and endurance	Inject service stalls and sensor faults. Complete a 24 h mixed-mode run with no unexplained reset, unbounded memory growth, or unreported data loss. Record maximum latency, heap/stack margins, mode current, and battery runtime.

Required delivery package

Deliver the schematic and PCB manufacturing files with the component list and power budget; the versioned ESP32-S3 firmware and isolated sensor bring-up environments; the BLE protocol and Python application; the backend/dashboard deployment configuration; and the calibration and integrated test records.

The release record identifies hardware revision, fitted module variants, firmware commit, dependency versions, board profile, configuration schema, and calibration revisions. Each goal G1-G7 is closed against the corresponding delivered hardware, software, and test evidence.

11 Design records and references

Installation settings and release records

Before commissioning, record the relay load and contact mapping; real alarm limits, reset rules, and maximum-on duration; measured sensor calibration coefficients and divider ratios; the charge and protection settings of the fitted battery modules; and the validated power-sense and transition timing. The first power prototype uses the drawn circuit; hardware hysteresis remains a later iteration.

Record the final S3 build configuration, GATT schema version, broker authentication, dashboard retention policy, and measured mode currents. These records accompany the release so that a future change to a sensor, module, or rule can be traced to its affected tests.

Project design references

[1] Power Circuit Diagram.pdf, supplied current drawing, page 1. Authority for the 3S battery, CN3303, HX-3S FL25A-A, MP1584 stages, IRLZ44N switching, isolation paths, and supply branches.

[2] Pin Layout.pdf, supplied current drawing, page 1. Authority for ESP32-S3-WROOM-1-N16R8, the sensor set, GPIO mapping, dividers, filters, USB, and relay-drive connections.

[3] Firmware Architecture.pdf, Revision 2, 4 October 2026. Source of task ownership, scheduling, copied records, rail epochs, handover, command handling, and timing targets. The current specification adapts these to the S3 hardware and SGP30; the older board pins and classic-ESP32 build assumptions are superseded.

[4] Redesign Specification.pdf, Revision 4, September 2026. Source of the project scope, custom PCB objective, BLE application, network dashboard, retained pH/TDS approach, and delivery requirements. Its earlier electrical configuration is superseded by [1] and [2].

Manufacturer and platform references

[5] Espressif. ESP32-S3-WROOM-1 and WROOM-1U datasheet v1.8.

[6] Sensirion. SGP30 datasheet v1.0, May 2020.

[7] Bosch Sensortec. BME280 datasheet.

[8] Analog Devices. DS18B20 datasheet.

[9] Espressif. ESP-IDF FreeRTOS guide for ESP32-S3.

[10] Monolithic Power Systems. MP1584 product specifications and datasheet.

[11] DFRobot. SEN0244 electrical specifications and compensation guidance.

Online references checked 16 October 2026. Platform API and power-management settings are matched to the versions recorded in the actual firmware build.